Second-Order Linear Homogeneous Equations

The auxiliary equation

1 Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.

2 Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.

Distinct real roots

3 Solve $y'' - 5y' + 6y = 0$.

4 Solve $y'' + 2y' - 3y = 0$, $y(0) = 1$, $y'(0) = 1$.

Repeated roots

5 Solve $y'' - 6y' + 9y = 0$.

6 Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.

Complex roots

7 Solve $y'' + 4y = 0$.

8 Solve $y'' - 2y' + 5y = 0$.

Application: a damped spring

9 A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).

10 Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.